

Kalman filter

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State estimation

Consider a linear time-invariant state-space model given by:

$$\dot{x} = Ax + Bu + v$$

$$y = Cx + w$$

where x is the state (vector), u is the input or control signal and y is the output signal, v is the process disturbance and w is measurement noise. The disturbance v and the noise w are **zero mean and Gaussian**

$$\mathbb{E}(v(s)v^T(t)) = R_v \delta(t - s)$$

$$\mathbb{E}(w(s)w^T(t)) = R_w \delta(t - s)$$

where δ is the unit impulse function (dirac function).

State estimation

The state estimator (observer) is given as:

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$$

The estimation error $\tilde{x} = x - \hat{x}$ can be computed as

$$\begin{aligned}\dot{\tilde{x}} &= \dot{x} - \dot{\hat{x}} = Ax + Bu + v - A\hat{x} - Bu - L(Cx + w - C\hat{x}) \\ &= (A - LC)\tilde{x} + v - Lw\end{aligned}$$

If $A - LC$ is stable, then the estimation error $\tilde{x}$ is a stationary stochastic process.

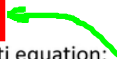
The covariance of the estimation error, $P_{\tilde{x}} = \mathbb{E}(\tilde{x}(t)\tilde{x}^T(t))$, is given by the following equation:

$$0 = (A - LC)P_{\tilde{x}} + P_{\tilde{x}}(A - LC)^T + R_v + LR_wL^T$$

The optimal observer minimizes $P_{\tilde{x}}$.

State estimation

The optimal observer gain, if the system is *observable*, is:

$$L = P_{\tilde{x}} C^T R_w^{-1}$$


where $P_{\tilde{x}} = P_{\tilde{x}}^T \geq 0$ is the solution to the Riccati equation:

$$0 = AP_{\tilde{x}} + P_{\tilde{x}}A^T + R_v - P_{\tilde{x}}C^T R_w^{-1} CP_{\tilde{x}}$$

The observer is called the **Kalman-Bucy filter**.

Similarities with LQR:

LQR $\leftrightarrow$ Kalman

$$A \leftrightarrow A^T \quad B \leftrightarrow C^T$$

$$S \leftrightarrow P \quad K \leftrightarrow L^T$$

$$Q_x \leftrightarrow R_v \quad Q_u \leftrightarrow R_w$$

The Kalman-Bucy filter is:

- always stable.
- *the optimal linear filter* for state estimation
- R_v and R_w are regarded as the design parameters.

Revisit Example - Vehicle steering (Ex 7.4)

Consider the following system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 12 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 6 \\ 3 \end{bmatrix} u + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

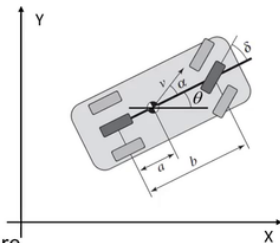
$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + w$$

where x_1 is the lateral position Y , x_2 is the heading orientation θ and u is the steering angle δ .

The process disturbance and the measurement noise are zero mean with covariance

$$R_v = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad R_w = \rho$$

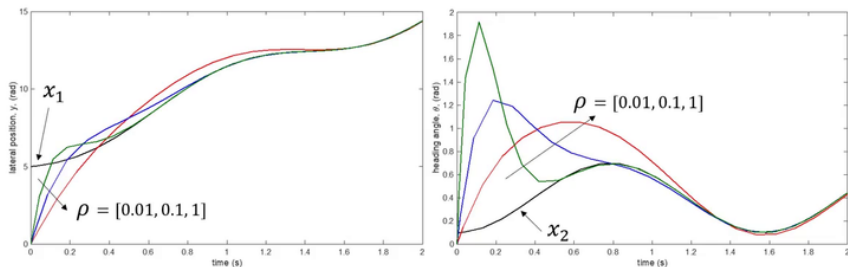
Design a Kalman filter to estimate the vehicle's states, from measurement of the lateral position.



Vehicle data: $v_0 = 12 \text{ m/s}$
 $a = 2 \text{ m}$
 $b = 4 \text{ m}$

Revisit Example - Vehicle steering (Ex 7.4)

Simulations using a sinusoidal input, with $x(0) = (5, 0.1)$ and $\hat{x}(0) = (0, 0)$:



State estimation – discrete time case

The covariance of the estimation error, $P_{\tilde{x}}[k] = \mathbb{E}(\tilde{x}[k]\tilde{x}^T[k])$, is given by:

$$P_{\tilde{x}}[k+1] = (A - L[k]C) P_{\tilde{x}}[k] (A - L[k]C)^T + R_v + L[k]R_w L^T[k]$$

The observer gain that minimizes $P_{\tilde{x}}[k]$ is given by

$$L[k] = A P_{\tilde{x}}[k] C^T (R_w + C P_{\tilde{x}}[k] C^T)^{-1}$$

This is the **discrete time Kalman filter**.

Note, that the Kalman filter is a recursive filter.

If $P_{\tilde{x}}[k]$ converges, then L is constant.

Predict

Predicted (*a priori*) state estimate

$$\hat{\mathbf{x}}_{k|k-1} = \mathbf{F}_k \hat{\mathbf{x}}_{k-1|k-1} + \mathbf{B}_k \mathbf{u}_k$$

Predicted (*a priori*) estimate covariance

$$\mathbf{P}_{k|k-1} = \mathbf{F}_k \mathbf{P}_{k-1|k-1} \mathbf{F}_k^\top + \mathbf{Q}_k$$

Update

Innovation or measurement pre-fit residual

$$\tilde{\mathbf{y}}_k = \mathbf{z}_k - \mathbf{H}_k \hat{\mathbf{x}}_{k|k-1}$$

Innovation (or pre-fit residual) covariance

$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^\top + \mathbf{R}_k$$

Optimal Kalman gain

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^\top \mathbf{S}_k^{-1}$$

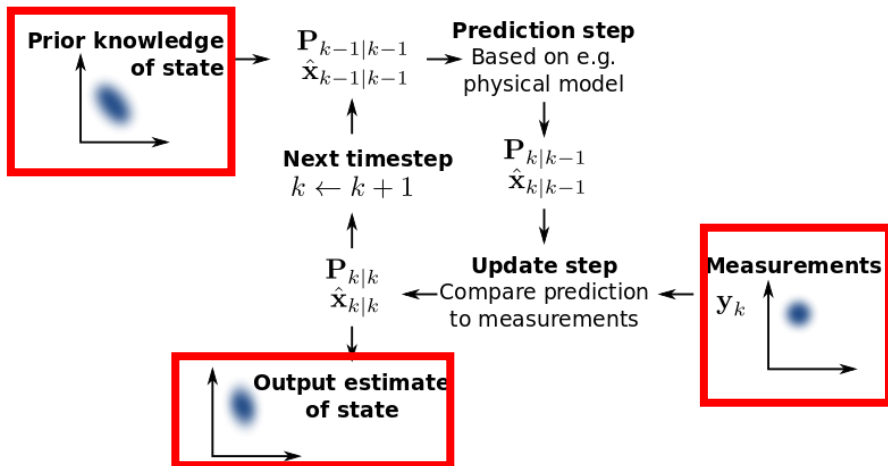
Updated (*a posteriori*) state estimate

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \tilde{\mathbf{y}}_k$$

Updated (*a posteriori*) estimate covariance

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_{k|k-1}$$

Discrete Kalman Filter Algorithm



- Karl J. Astrom and Richard M. Murray *Feedback Systems*. Version v3.0i. Princeton University Press. September 2018. Chapter 8.
- Wikipedia. Kalman filter. https://en.wikipedia.org/wiki/Kalman_filter.